TECHNICAL BENCHMARK GUIDE

SpeedCloud

Performance benchmark report explained

A practical reading guide for the report's cost, storage, compute, network, and database measurements - including what the numbers mean and how to validate them on a real workload.



Source visual reproduced from SpeedCloud_Performance_Benchmark_Report-May2026.pdf. The report's results are treated here as reported benchmark claims and are not independently re-run.

01 | EXECUTIVE SUMMARY

What the benchmark report is claiming

The report compares SpeedCloud with AWS, Google Cloud Platform, and Microsoft Azure using equivalent 8 vCPU, 16 GB RAM, and 100 GB instance configurations. It covers compute with Geekbench 6, storage with FIO, network with iPerf3, and database throughput with Sysbench.



Figure 1. Summary claims and test scope from the source report.

Area	Reported headline	How to read it
Cost	Up to 50%+ savings versus global cloud providers on equivalent list-price instance configurations.	This is a price comparison, not automatically a complete TCO comparison. Include data movement, support, licensing, operations, and exit effort.
Storage	Up to 741% better sequential-read IOPS versus AWS in the report.	IOPS is useful only with the same block size, queue depth, durability, read/write mix, and latency target as the application.
Compute	Up to 179% better multi-core Geekbench score versus Azure in the report.	A synthetic CPU score is directional; validate application throughput, tail latency, and cost per completed job.
Network	30.8 Gbit/s sender and 30.7 Gbit/s receiver for SpeedCloud on iPerf3.	Confirm topology, MTU, concurrency, placement, encryption, and whether the workload is east-west or internet-facing.

Benchmark boundary

- The report says the tests were conducted in January 2025, while the document is dated May 2026.
- The compared configurations are described as equivalent, but equivalence should be confirmed for CPU generation, storage class, network path, and pricing assumptions.

02 | HOW TO READ THE METRICS

The unit and direction of a metric determine its meaning

Benchmark results can look impressive while still being a poor proxy for production. First identify the metric, its direction of goodness, and the workload behavior it represents.

Metric	What it measures	Higher or lower?	Production question
IOPS	Storage operations completed per second. FIO in the report uses 4K blocks and sequential or random read/write patterns.	Higher is generally better, but latency and durability also matter.	Does the application use the same block size, queue depth, and read/write mix?
Geekbench score	Synthetic CPU performance for single-core and multi-core execution.	Higher is better for the tested workload, but scores do not equal application throughput.	Is the application single-threaded, parallel, memory-bound, or license-limited?
Gbit/s	Network transfer bitrate from iPerf3 sender or receiver tests.	Higher is better for bandwidth-heavy paths; latency and packet loss still matter.	Can the application actually fill the path under its packet size and concurrency?
MiB/sec	Database file-I/O transfer rate in the Sysbench test.	Higher is better for throughput; it does not alone describe transaction latency.	
Max latency	Maximum latency reported by the Sysbench database test, in milliseconds.	Lower is better. Tail behavior matters more than an average alone.	What are p95/p99 latency and transaction error behavior under concurrency?

What can make a benchmark misleading

- A storage result can change with queue depth, filesystem, cache state, volume policy, and whether the test is local or network-attached.
- A network result can change with placement, congestion, TCP settings, MTU, encryption, and the number of parallel streams.
- A database result can change with engine version, buffer pool, schema, indexes, dataset size, concurrency, and durability settings.
- A cost result can change with commitment discounts, support plans, egress, backups, licenses, and the amount of engineering effort required.

03 | STORAGE PERFORMANCE

IOPS is valuable when the workload is I/O-bound

The source report uses FIO with a 4K block size and reports sequential and random read/write IOPS. These are useful indicators for storage-intensive systems, but they should be paired with latency, queue depth, durability, and cost measurements.



Figure 2. FIO storage results from the source report.

Workload pattern	SpeedCloud result	Reported comparison	Where it matters
Sequential read	785k IOPS	+741% versus AWS (93.3k)	Large scans, media pipelines, analytics reads, and sequential data movement.
Sequential write	199k IOPS	+1,740% versus Azure (10.8k)	Bulk ingest, log streams, staging, and sequential export.
Random read	124k IOPS	+3,333% versus Azure (3.6k)	Database lookups, indexes, metadata, and small-object access.
Random write	102k IOPS	+2,733% versus Azure (3.6k)	Transactional writes, journaling, and random update patterns.

Example: high-I/O database

- Run the application's actual read/write mix, block size, queue depth, concurrency, and durability mode.
- Record p95/p99 latency, IOPS, transaction rate, CPU wait, checkpoint behavior, and backup impact.
- Confirm that the tested storage is the same class and attachment model offered in production.

04 | COMPUTE AND NETWORK PERFORMANCE

Separate CPU capacity from network path capacity

The report combines Geekbench 6 compute results with iPerf3 sender and receiver results. A workload can benefit from one and not the other, so compute and network should be validated independently and then together.

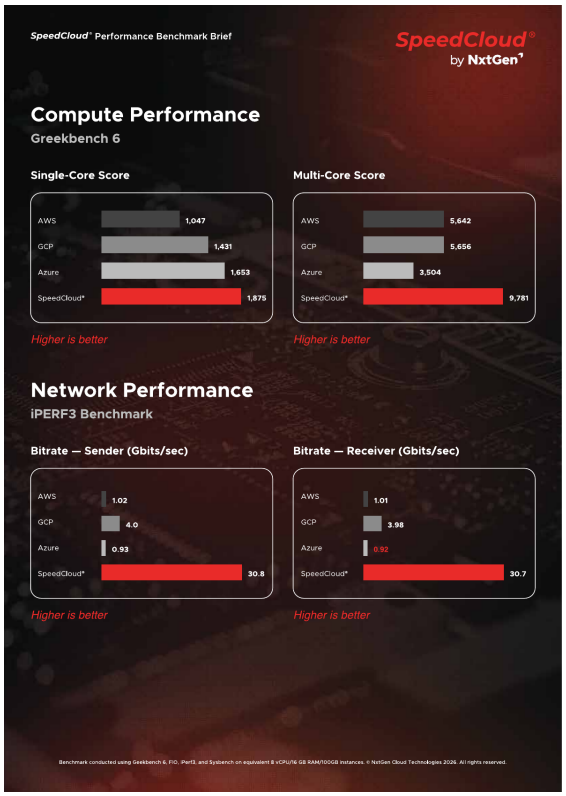


Figure 3. Geekbench 6 and iPerf3 results from the source report.

Test	SpeedCloud	Reference point	Interpretation
Single-core	1,875	Azure 1,653	Useful for latency-sensitive or lightly threaded work; test application response time.
Multi-core	9,781	Azure 3,504	Useful for parallel work; report states +179% versus Azure.
iPerf3 sender	30.8 Gbit/s	Azure 0.93 Gbit/s	Tests outbound path capacity under the report's network setup.
iPerf3 receiver	30.7 Gbit/s	Azure 0.92 Gbit/s	Tests inbound path capacity; validate application-level transfer and packet behavior.

Example workload mapping

- API service: prioritize single-thread behavior, p95/p99 latency, load-balancing behavior, and error rate.
- AI/HPC, analytics, replication, or backup: validate multi-core throughput, node-to-node bandwidth, sustained transfer, packet behavior, and job or recovery completion.

05 | DATABASE THROUGHPUT

Read transfer rate and latency together

The Sysbench page reports database file-I/O transfer rate and maximum latency. The source shows SpeedCloud at 4,662 MiB/sec transfer rate and 0.1 ms maximum latency, equal to GCP on the latter metric in the summary table.

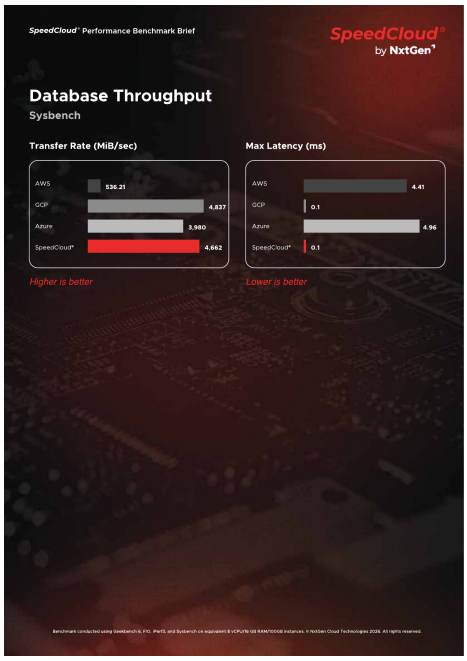


Figure 4. Sysbench database-throughput results from the source report.

Provider	Transfer rate (MiB/sec)	Max latency (ms)	How to validate
AWS	536.21	4.41	Confirm engine, storage, dataset, and durability settings before comparison.
GCP	4,837	0.10	The report shows higher transfer rate than SpeedCloud but similar maximum latency.
Azure	3,980	4.96	Compare tail latency and transaction behavior, not only the throughput number.
SpeedCloud	4,662	0.10	Test production schema, concurrency, backup, failover, and cost per transaction.

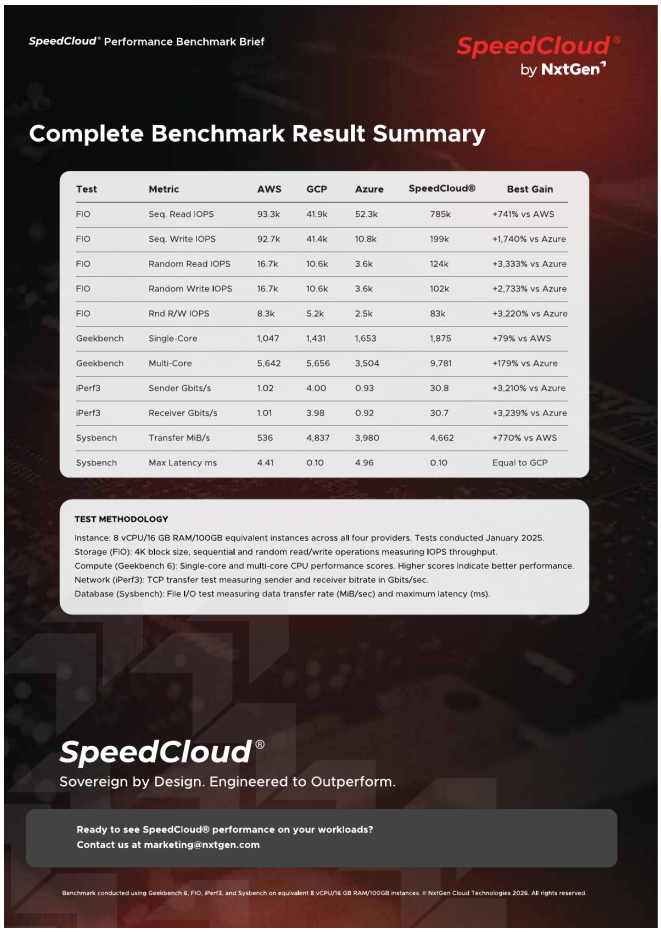
Database test questions

- Which database engine and version were used, and does the target workload use the same engine behavior?
- What are p95 and p99 transaction latency, throughput under concurrency, lock waits, and error rate?
- How do backup, replication, encryption, maintenance, and failover affect the result?
- Measure cost per transaction or completed job, not only raw MiB/sec.

06 | COMPLETE RESULT SUMMARY

Use the report's table as a map, not as the final decision

The source report consolidates FIO, Geekbench, iPerf3, and Sysbench into one table. Reproducing the complete summary is useful for orientation; the engineering decision still depends on workload-specific testing and a transparent methodology.



Test	Metric	AWS	GCP	Azure	SpeedCloud®	Best Gain
FIO	Seq. Read IOPS	93.3k	41.9k	52.3k	785k	+741% vs AWS
FIO	Seq. Write IOPS	92.7k	41.4k	10.8k	199k	+1,740% vs Azure
FIO	Random Read IOPS	16.7k	10.6k	3.6k	124k	+3,333% vs Azure
FIO	Random Write IOPS	16.7k	10.6k	3.6k	102k	+2,733% vs Azure
FIO	Rnd R/W IOPS	8.3k	5.2k	2.5k	83k	+3,220% vs Azure
Geekbench	Single-Core	1,047	1,431	1,653	1,875	+79% vs AWS
Geekbench	Multi-Core	5,642	5,656	3,504	9,781	+179% vs Azure
iPerf3	Sender Gbits/s	1.02	4.00	0.93	30.8	+3,210% vs Azure
iPerf3	Receiver Gbits/s	1.01	3.98	0.92	30.7	+3,239% vs Azure
Sysbench	Transfer MiB/s	536	4,837	3,980	4,662	+770% vs AWS
Sysbench	Max Latency ms	4.41	0.10	4.96	0.10	Equal to GCP

Figure 5. Complete benchmark summary and methodology from the source report.

Questions to ask before using a headline result

- Were the instance sizes truly comparable in CPU generation, storage attachment, network placement, and service tier?
- Are the results repeatable across time, zones, concurrency levels, and multiple runs?
- Were caches warm or cold, and were background workloads or noisy-neighbor effects controlled?
- Does the customer's application show the same bottleneck as the benchmark?
- What are the complete costs and customer responsibilities around the tested service?

07 | WORKLOAD VALIDATION

Turn benchmark claims into acceptance criteria

A benchmark is useful when it gives the team a measurable hypothesis. The proof of concept should reproduce the relevant behavior, test failure modes, and compare the complete operating model.

Workload	Metrics to capture	Acceptance example
Transactional database	p95/p99 latency, transaction rate, IOPS, lock waits, backup impact, failover time.	Meets latency and recovery targets at peak concurrency with agreed cost per transaction.
Data pipeline or analytics	Sequential throughput, memory/CPU utilization, network transfer, job completion, cost per job.	Completes the daily pipeline within the window without saturating storage or network.
API or web service	p95/p99 response time, error rate, requests/sec, scale-out time, cache hit ratio.	Maintains the target latency and error budget during peak and failure scenarios.
Backup or replication	Sustained transfer, encryption overhead, RPO, RTO, restore time, data integrity.	Restores within the agreed RTO and meets RPO with verified data integrity.

Recommended proof-of-concept sequence

- Baseline the current platform with the same workload, dataset, concurrency, and measurement window.
- Deploy an equivalent configuration and document every deviation from the source report's test setup.
- Run normal, peak, failure, backup/restore, maintenance, and scaling scenarios.
- Compare technical results, operator effort, support model, full cost, portability, and exit effort.

The report itself says the tests used equivalent 8 vCPU/16 GB RAM/100 GB instances. Treat that as the starting configuration, not a universal recommendation.

08 | DECISION CHECKLIST

Questions to close before relying on the results

Use this checklist with infrastructure, application, database, security, finance, and operations teams before converting a benchmark headline into a production choice.

Area	Questions to answer
Methodology	Are CPU generation, storage class, network path, dataset, engine version, queue depth, concurrency, and run count comparable?
Metric fit	Which reported metric maps directly to the application's bottleneck, and which production metric will prove improvement?
Performance	Have p95/p99 latency, throughput, saturation, error rate, and failure behavior been measured under expected peak load?
Data	Are durability, consistency, backup, replication, encryption, retention, and restore requirements included?
Network	Are placement, path type, MTU, packet loss, encryption, bandwidth, and east-west versus north-south traffic understood?
Cost	Does the comparison include list price or discount assumptions, egress, support, licenses, operations, migration, backup, and exit cost?
Acceptance	Are the benchmark setup, raw results, scripts, dates, and responsibility matrix documented and reviewable?

Technical conclusion

The benchmark report presents strong comparative results for the tested configurations, especially in storage IOPS, multi-core compute, and iPerf3 network throughput. The sound technical conclusion is narrower: SpeedCloud may be a strong candidate for workloads whose bottlenecks resemble the tested patterns, provided the results are reproduced with the customer's data, topology, service settings, and complete cost model.

All benchmark values, pricing assumptions, test dates, product scope, service limits, and performance claims should be confirmed against the current product documentation and a signed commercial agreement.